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Box M-test

there are 2 independent populations $N_p(\mu_1, \Sigma_1)$, $N_p(\mu_2, \Sigma_2)$

$x_1, \dots, x_{n_1}$ are randomly independently drawn from $N_p(\mu_1, \Sigma_1)$

$y_1, \dots, y_{n_2}$ $\parallel$ $N_p(\mu_2, \Sigma_2)$

$$H_0: \Sigma_1 = \Sigma_2$$

$$H_1: \Sigma_1 \neq \Sigma_2$$

$$S_{\text{pooled}} = \frac{(n_1-1)S_1 + (n_2-1)S_2}{n_1 + n_2 - 2}$$

Box M-statistic

Define M as :

$$M = (n_1-1) \ln |S_1| + (n_2-1) \ln |S_2| - (n_1+n_2-2) \ln |S_{\text{pooled}}|$$

Correction factor C :

$$(1-C)M \stackrel{d}{\sim} \chi^2_{\nu} \quad \text{asymptotically, when sample size is large}$$

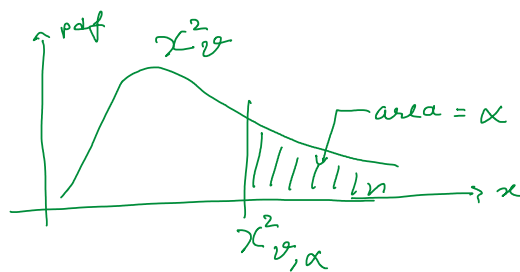
$$\text{here } \nu = \frac{p(p+1)}{2} \quad [p = \text{number of features}]$$

$$g = 2 \quad (\text{number of groups in our context})$$

$$C = \frac{2p^2 + 3p - 1}{6(p-1)(g-1)} \left(\frac{1}{n_1-1} + \frac{1}{n_2-1} - \frac{1}{n_1+n_2-2} \right)$$

Reject H_0 at α -level of significance if :

$$(1-C)M > \chi^2_{\nu, \alpha}$$

Rough proof:

for g : number of populations ($g \geq 1$)

populations are $N_p(\mu_i, \Sigma_i)$ ($1 \leq i \leq g$) [all independent populations]

$$H_0: \Sigma_1 = \dots = \Sigma_g$$

H_1 : Not all Σ 's are equal

$$x_{i1}, \dots, x_{in_i} \sim N_p(\mu_i, \Sigma_i) \quad [\text{randomly chosen}]$$

$\bar{x}_i$, S_i $\rightarrow$ sample mean and sample cov Mat.

$$\text{Define } A_i = \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)(x_{ij} - \bar{x}_i)^T = (n_i-1)S_i$$

$$\text{MLE for } \hat{\mu}_i = \bar{x}_i$$

$$\text{for } \hat{\Sigma}_i = \frac{A_i}{n_i} = \frac{(n_i-1)}{n_i} S_i$$

Recall from earlier notes (log Likelihood)

$$\log L_i(\mu_i, \Sigma_i) = -\frac{n_i}{2} \ln |\Sigma_i| - \frac{1}{2} \sum_{j=1}^{n_i} (x_{ij} - \mu_i)^T \Sigma_i^{-1} (x_{ij} - \mu_i) + \text{constant}$$

↑
does not matter
will cancel out
[due to 2π factor]

To find MLE $\hat{\mu}_i = \bar{x}_i$

$$\log L_i(\hat{\mu}_i, \Sigma_i) = -\frac{n_i}{2} \ln |\Sigma_i| - \frac{1}{2} \text{trace}(\Sigma_i^{-1} A_i) + \text{constant}$$

$$\log L_i(\hat{\mu}_i, \hat{\Sigma}_i) = -\frac{n_i}{2} \ln \left| \frac{A_i}{n_i} \right| - \frac{n_i p}{2} + \text{constant}$$

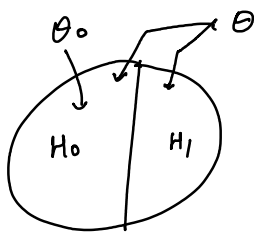
under H_0 :

$$\log L_i(\hat{\mu}, \hat{\Sigma})^{H_0} = -\frac{N}{2} \ln \left| \frac{A}{N} \right| - \frac{Np}{2} + \text{constant}$$

$$N = \sum_{i=1}^g n_i \quad A = \sum_{i=1}^g A_i$$

Now Likelihood ratio test (LRT):

$$\hat{\Lambda}_{(\mu, \Sigma)} = \frac{\sup_{(\mu, \Sigma) \in \Theta_0} L(\mu, \Sigma)}{\sup_{(\mu, \Sigma) \in \Theta} L(\mu, \Sigma)}$$



$$= \frac{L(\hat{\mu}, \hat{\Sigma})^{H_0}}{L(\hat{\mu}, \hat{\Sigma})} \quad \left. \vphantom{\frac{L(\hat{\mu}, \hat{\Sigma})^{H_0}}{L(\hat{\mu}, \hat{\Sigma})}} \right\} \begin{array}{l} \text{convert ratio} \\ \text{to log L} \end{array}$$

$$= \exp \left(\log L(\hat{\mu}, \hat{\Sigma})^{H_0} - \sum_{i=1}^g \log L(\hat{\mu}_i, \hat{\Sigma}_i) \right)$$

$$\therefore -2 \ln \hat{\Lambda} = -2 \log L(\hat{\mu}, \hat{\Sigma})^{H_0} + 2 \sum_{i=1}^g \log L(\hat{\mu}_i, \hat{\Sigma}_i)$$

$\uparrow$ derived $\uparrow$ summation because independent

$$= N \ln \left| \frac{A}{N} \right| - \sum_{i=1}^g n_i \ln \left| \frac{A_i}{n_i} \right|$$

and note the constants will cancel out

and $2 \ln \hat{\Lambda} \stackrel{d}{\sim} \chi^2_{\nu}$ $\nu = \dim \Theta - \dim \Theta_0$

this is then related to M

$$\nu \simeq \frac{p(p+1)}{2}$$